Práctica tres: Modelado y control del sistema musculoesqueletico

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

Tecnológico Nacional de México [TecNM - Tijuana], Blvd. Alberto Limón Padilla s/n, C.P. 22454, Tijuana, B.C., México

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Información general



Nombre del alumno: **Ibarra Monroy Ivana Valeria**

Número de control: 23210709

Correo institucional: l23210709@**tectijuana.edu.mx**

Asignatura: **Modelado de Sistemas Fisiológicos**

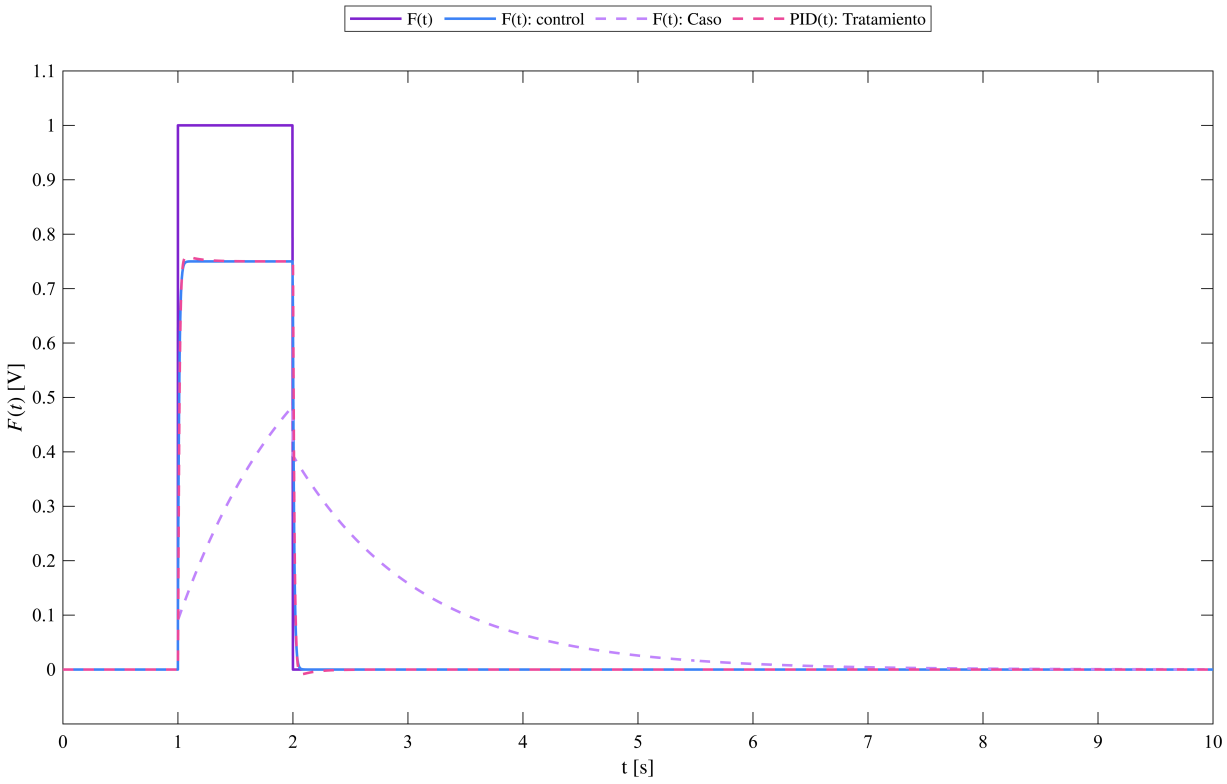
Docente: **Dr. Paul Antonio Valle Trujillo; paul.valle@tectijuana.edu.mx**

Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('ibarra2321070903')
parameters.StopTime = '10';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-2';
```

Respuesta: impulso unitario

```
x = sim('ibarra2321070903',parameters);
plotsignals(x.t,x.F,x.Fsx,x.Fsy,x.PID, 'impulso unitario')
```



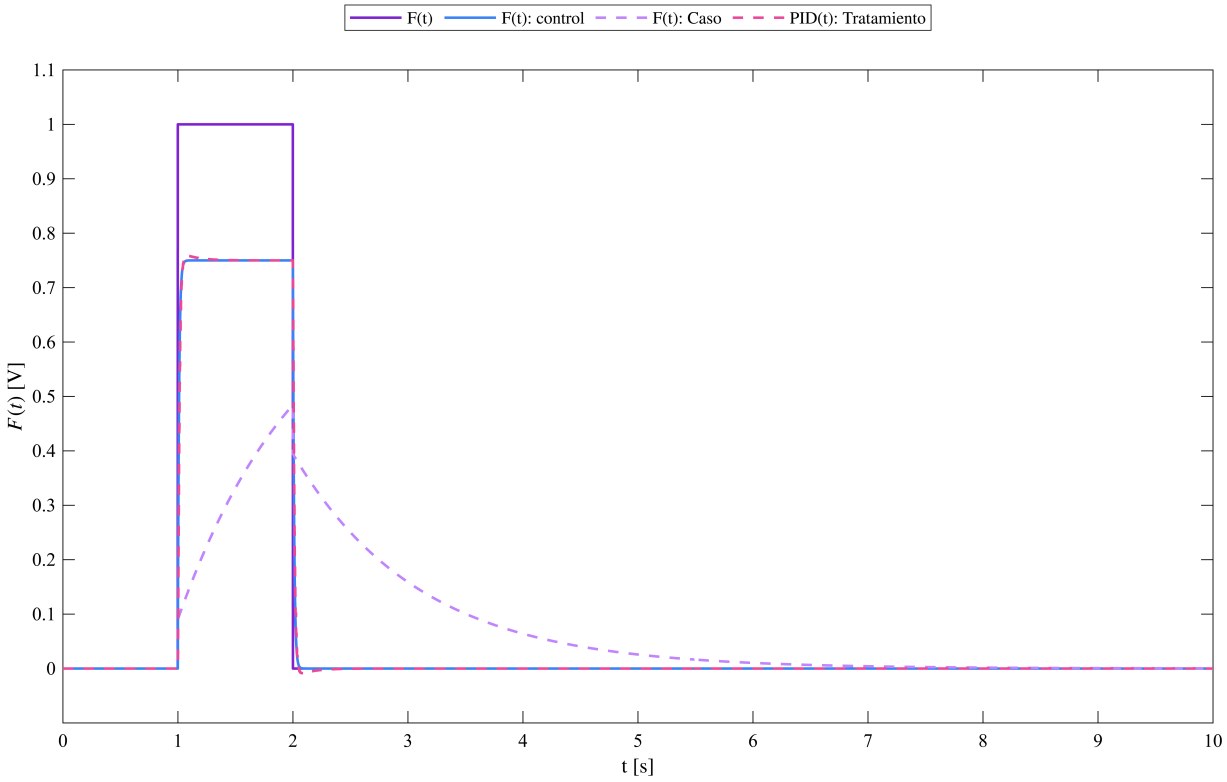
Modelo de EDOs

```
dt = 1e-3;
t = (0:dt:10)';
F = zeros(length(t),1); F(1/dt:2/dt) = 1;
PID = interp1(x.t, x.PID, t, 'previous');

par_control = [0.25,10e-6,100e-6,100];
par_caso = [0.25,10e-6,100e-6, 10e3];

Fs_control = musculoesqueletico(par_control,F,t);
Fs_caso = musculoesqueletico(par_caso,F,t);

plotsignals(t,F,Fs_control,Fs_caso,PID, 'EDOsys')
```




```
function Fs = musculoesqueletico(par,F, t)
alpha = par(1); Cs = par(2); Cp = par(3); R = par(4); dt = t(2);
n = round(t(end)/dt);
x = zeros(length(t), 1); x(1) = 0;
Fs = zeros(length(t),1); Fs(1) = 0;
for i = 1:n
    x(i+1) = x(i) + ((F(i)-x(i))/(R*(Cp+Cs)))*dt; %EDO
    Fs(i+1) = Cs*F(i)/(Cp+Cs) + (1 - alpha - Cs/(Cp+Cs))*x(i); %Función
end
end
```

Funcion: Respuestas de las senales

```
function plotsignals(t,F,Fsx,Fsy,PID,signal)
set(figure(),'color','w')
set(gcf,'units','inches','position', [0.5,0.5,12,7])
set(gca, 'fontname','times new roman','fontsize',11)
hold on; grid off; box on;
clr1 = [126, 34, 206]/255;
clr2 = [59, 130, 246]/255;
clr3 = [192, 132, 252]/255;
clr4 = [236, 72, 153]/255;
order = [clr1; clr2; clr3;clr4]; colororder(order)
xlabel('t [s]','Interpreter','latex')
```

```
ylabel('$F(t)$ [V]', 'Interpreter','latex')
xlim([0 10]); xticks(0:1:10)
ylim([-0.1 1.1]); yticks(0:0.1:1.1)
if signal == "impulso unitario"

plot(t,F,'-',t,Fsx,'-',t,Fsy,'--',t,PID,'--','LineWidth',1.5,'MarkerSize',5,'
MarkerIndices',...
        1:1e3:length(t));

    L = legend('F(t)','F(t): control','F(t): Caso','PID(t): Tratamiento');

set(L,'interpreter','latex','location','northoutside','orientation','horizont
al')

elseif signal == "EDOsys"

plot(t,F,'-',t,Fsx,'-',t,Fsy,'--',t,PID,'--','LineWidth',1.5,'MarkerSize',5,'
MarkerIndices',...
        1:1e3:length(t));

    L = legend('F(t)','F(t): control','F(t): Caso','PID(t): Tratamiento');

set(L,'interpreter','latex','location','northoutside','orientation','horizont
al')
end
set(gcf,'Color','white')
set(gca,'Color','white','XColor','black','YColor','black')
set(L,'Color','white','TextColor','black')
exportgraphics(gcf,'Sistemamusc_matlab.pdf','ContentType','vector')
exportgraphics(gcf,'edosys.pdf','ContentType','vector')
end
```